What a prime-programmed polymer can read: sieve composition, congruence traps, and arithmetic blindness

Ruqing Chen*
GUT Geoservice Inc., Montreal, Quebec, Canada
(Dated: October 1, 2026)

A self-avoiding chain whose bending stiffness follows the gaps of a set of primes (the prime-driven topological polymer, PDTP) registers the ordering of the prime gaps, but earlier work showed that this imprint is carried by the smooth prime-number-theorem envelope at the small-prime end of the chain. Here we ask whether the fold can register anything finer: the congruence structure that confines twin and Sophie Germain primes to 2 mod 3 and odd Landau primes n^2+1 to 1 mod 4, which in localized Bost–Connes systems decides whether the equilibrium state is unique. We answer in five tests. (i) A theorem: any order contrast against a multiset-preserving surrogate is blind to every congruence channel locked by the gap multiset, which includes all single-class confinements. (ii) In a prime window at 10^8, where the envelope is flat, the Langevin polymer does not distinguish the primes from ten fixed shuffles at the sequence level ($Z_{\mathrm{seq}} = +1.04$, $p = 0.33$). (iii) The exact positional readout of the ideal chain does not respond to jitter surrogates that null or preserve a single mod-q channel (contrasts ≤ 0.004 against a shuffle spread of 0.02). (iv) The exact scattering function does not either; its apparent response is an artefact of the jitter, which changes the gap multiset, and we show that no jitter pair can avoid this. (v) What remains is composition. Against Granville's sieve models drawn with the family's own density envelope and anchor count, the stiffness composition of every family converges monotonically with sieve depth z; the single-class confinement is the $z = 3$ layer of that convergence. The kernel-versus-generator dichotomy that separates Landau from twin primes in the Bost–Connes setting has no polymer counterpart: twin primes lie further from their sieve model than Landau primes do, because a congruence condition affects the polymer in proportion to the ratio of modulus to mean gap. The polymer reads arithmetic only through the positional envelope and sieve-depth composition.

I. INTRODUCTION

The prime-driven topological polymer (PDTP) maps the gap between consecutive anchors of an integer set onto the local bending stiffness of a coarse-grained self-avoiding chain [1]. For the primes, the collapsed chain is measurably less radially sorted than a chain built from a random permutation of the same gaps, at every size and after complete relaxation [1, 2]. Two results fix what that imprint is. An exact solution of the ideal chain shows that any readout carried by chain-scale composition produces the size-proportional correlation length observed for it, and surrogates that keep only the smooth prime-number-theorem (PNT) envelope reproduce it [2]. Restricting the readout to bead subsets shows that the imprint sits at the small-prime end of the chain, where the PNT drift is steepest [1].

That leaves a sharper question. The primes carry arithmetic structure finer than the envelope: congruence conditions. Some prime families are confined to a single residue class — twin and Sophie Germain primes to 2 mod 3, odd Landau primes n^2+1 to 1 mod 4 — and in the statistical mechanics of localized Bost–Connes systems this confinement matters decisively [3, 4]. Confinement to a class that lies in the kernel of a Dirichlet character (Landau) makes the equilibrium state non-unique at every temperature; confinement to a class that generates the unit group (twin, Sophie Germain) obstructs nothing [3, Ch. 37]. Does a polymer whose stiffness is programmed by these families register the congruence structure, and does it distinguish the two kinds of confinement?

We find that it does not, and we show why. Section III proves that order contrasts are blind to congruence channels locked by the gap multiset. Sections IV–VI test the remaining routes: a drift-free window, the positional readout, and the scattering function. Section VII shows that composition is the only channel through which the congruence structure reaches the fold, measures it against sieve models, and answers the kernel-versus-generator question. Section VIII relates the result to the blindness of tensorizing invariants in the Bost–Connes setting [3, Ch. 4].

II. MODEL, FAMILIES, AND SURROGATES

Beads $i = 0, \ldots, N-1$ are identified with the integers. Given a sorted anchor set A, a bead between consecutive anchors $a < i < a'$ carries the gap $g_i = a' - a$ and an anchor bead carries $g_i = 0$; the stiffness is

$$k_i = \begin{cases} \min[k_{\mathrm{hinge}} e^{\alpha g_i},\, k_{\max}], & i \notin A, \\ k_{\mathrm{hinge}}, & i \in A, \end{cases} \tag{1}$$

with $k_{\mathrm{hinge}} = 0.25$ and $k_{\max} = 500$ [1]. To compare families of very different density on an equal footing we

* ruqing@hotmail.com; Archived at doi:10.5281/zenodo.23094095.

set $\alpha = 0.15 \times 9.52/\bar{g}$, so that $\alpha\bar{g}$ equals its value for the primes at $N = 4 \times 10^4$.

Readouts. The exact ideal chain with bending energy $k_i(1 - \cos\theta_i)$ has tangent correlations $\langle \mathbf{t}_a \cdot \mathbf{t}_b \rangle = \prod_{k=a+1}^{b} \mathcal{L}(k_k)$, with $\mathcal{L}$ the Langevin function and $\beta = 1$ [1]. From this we obtain, exactly, the radius of gyration R_g^2; the characteristic ratio $C_N = \langle R_{ee}^2 \rangle/(N-1)$; the mean bond cosine $\langle \mathcal{L}(k) \rangle$; the curvature energy per bead $\langle k(1 - \mathcal{L}(k)) \rangle$; the positional readout $\rho_{id} = \mathrm{Spearman}(k_i, \langle |\mathbf{r}_i - \mathbf{r}_{cm}|^2 \rangle)$, computed in $O(N)$; and the scattering function in the Gaussian closure, $S(q) = N^{-1} \sum_{ij} \exp(-q^2 \langle R_{ij}^2 \rangle/6)$, computed in $O(N^2)$. The Langevin model of Sec. IV adds excluded volume and annealing [1, 5, 6].

Families. Anchors below N are the primes; the twin primes (smaller member p with $p+2$ prime); the Sophie Germain primes ($2p+1$ prime); and the Landau primes n^2+1. For $p > 3$ twin and Sophie Germain primes lie in 2 mod 3, hence in 5 mod 6, so their gaps lie in $6\mathbb{Z}$; odd Landau primes lie in 1 mod 4, so their gaps lie in $4\mathbb{Z}$ [3, Ch. 37, Lemma 2.1]. Our counts to 2×10^5 reproduce the table of Ref. [3, Ch. 37] exactly (2159, 2056 and 63 members).

Surrogates. S1 permutes the gaps (exact multiset, order destroyed). The sieve model S_7-G(z) is Granville's refinement of Cramér's model [7, 8], adapted to each family: its candidates are the integers that pass the family's own sieve conditions at depth z (primes: n; twin: n and $n+2$; Sophie Germain: n and $2n+1$; Landau: n^2+1), each coprime to $\prod_{p \le z} p$ or itself a prime $\le z$. We draw it bin by bin: bins hold 20 consecutive family members and each bin receives the family's own count. This holds the anchor count and the smooth density envelope fixed, so that S_7-G(z) differs from the family only in arithmetic, and z orders the arms by arithmetic depth. $z = 1$ imposes no sieve; for Landau it keeps the polynomial support n^2+1. The Poisson arm draws from all integers with the same binning. Finally, the jitter pair of Sec. V shifts every interior anchor by a random integer and re-sorts, keeping both end anchors fixed and every shifted anchor between them: the *kill* arm by $\varepsilon \sim \mathrm{U}\{0, \ldots, q-1\} - \lfloor (q-1)/2 \rfloor$, which nulls every Bragg channel a/q' with $q' \mid q$; the *keep* arm by $q\,b$, $b \sim \mathrm{U}\{0, 1\}$, which preserves them exactly. Section VI adds a variance-matched keep arm.

III. ORDER BLINDNESS TO LOCKED CHANNELS

Write the Bragg amplitude of an anchor set at rational frequency a/q as $B_{a/q}(A) = |A|^{-1} \sum_{x \in A} e^{2\pi i x a/q}$.

Theorem 1 (Order blindness)**.** *Let all gaps of A lie in $q\mathbb{Z}$. Then every multiset-preserving reordering A' of the gaps with the same first anchor satisfies $|B_{a/q'}(A')| = |B_{a/q'}(A)| = 1$ for every $q' \mid q$. In particular, for any readout $\mathcal{R}$, the contrast between the true family and a multiset-preserving surrogate class carries no information about the confinement of the family to one residue class modulo q.*

Proof. If every gap is a multiple of q, every anchor is congruent to the first anchor modulo q, for any order of the gaps. All terms of $B_{a/q'}$ then have the same phase, and the amplitude has modulus one in both arms. □

The fixed first anchor is not needed. If a reordering starts from a different anchor c, every anchor is congruent to c modulo q, and the amplitude is the overall phase $e^{2\pi i c a/q'}$, of modulus one. The statement is therefore invariant under any translation of the window.

Corollary 1. *For twin and Sophie Germain primes ($q = 6$) and for odd Landau primes ($q = 4$) the single-class confinement is invisible to every True-versus-shuffle contrast, including all those of Refs. [1, 2].*

The theorem closes the route that earlier surrogate hierarchies relied on: ordering contrasts can detect structure in how the gaps are arranged, but a confinement to one class is not an arrangement. It is a property of the gap values, that is, of composition. The remaining sections test whether the fold registers congruence structure by any other route.

TABLE I. Drift-free window $[10^8, 10^8 + 10^4)$, Langevin model. Upper block: arms with a fresh surrogate sequence per trajectory, $n = 40$, mean ± s.e.m. Lower block: one-way analysis of the fixed-shuffle class and the sequence-level contrast with the true chain.

arm	ρ	Welch p vs true
true primes	$+0.2429 \pm 0.0075$	—
S1 (fresh)	$+0.2108 \pm 0.0076$	0.004
S7 (IAAFT) [9]	$+0.2196 \pm 0.0080$	0.038
S_7-G(5)	$+0.2450 \pm 0.0097$	0.87
S_7-G(5) shuffled, run-paired	$+0.2398 \pm 0.0087$	0.79
fixed S1 class, $K \times m$	10×5	
$\hat{\mu}$, $\hat{\sigma}_{seq}$, $\hat{\sigma}_{th}$	+0.2241, 0.0143, 0.0477	
true $-\hat{\mu}$, Z_{seq} (p)	+0.019, +1.04 (0.33)	
sequence means above true	3 of 10	

IV. A DRIFT-FREE WINDOW

If the ordering imprint of the primes contained anything beyond the PNT envelope, it would survive where the envelope is flat. In the window $[10^8, 10^8 + 10^4)$ the relative change of $\ln i$ across the chain is 5×10^{-6}. We folded its 551 primes with the Langevin model of Ref. [1] (5000 annealing steps; α rescaled to 0.0806 to keep the stiffness contrast of the origin chain) and compared them with shuffles at the sequence level: $K = 10$ fixed shuffle sequences with $m = 5$ trajectories each (Table I, Fig. 1).

The naive Welch test against fresh shuffles gives $p = 0.004$. It is anti-conservative. The between-sequence

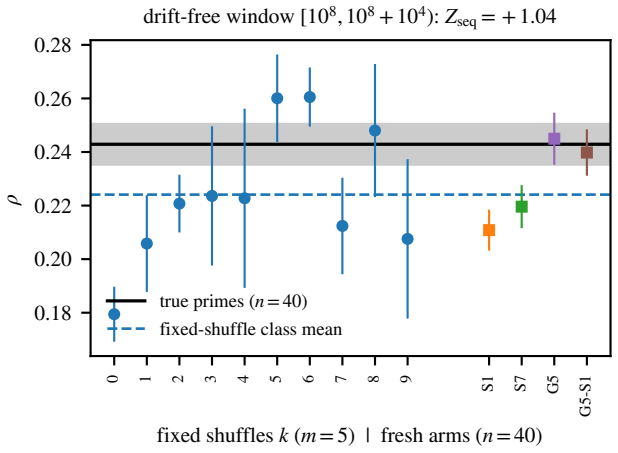


FIG. 1. Drift-free window. Dots: the ten fixed shuffle sequences (mean ± s.e.m. of five trajectories each); dashed line: their class mean. Black line and band: the true primes. Squares: fresh-sequence arms (S1, IAAFT, sieve model $z = 5$ and its run-paired shuffle).

TABLE II. Exact positional readout ρ_{id} at $N = 4 \times 10^4$, 40 realizations per arm. n: anchors (SG: Sophie Germain). z: true family against the S1 ensemble. Δ: mean change of ρ_{id} under the kill and keep jitter arms relative to the true family.

			$q=3$		$q=6$	
family	n	z	Δ_{kill}	Δ_{keep}	Δ_{kill}	Δ_{keep}
primes	4203	−2.4	+0.0007	+0.0016	+0.0015	+0.0037
twin	591	0.0	+0.0001	−0.0005	+0.0004	+0.0001
SG	565	0.0	+0.0001	+0.0004	0.0000	+0.0013
Landau	34	−1.4	0.0000	−0.0001	0.0000	+0.0001

standard deviation of the shuffle class, $\hat{\sigma}_{\mathrm{seq}} = 0.014$, is of the order of the effect itself, and it enters the variance of the contrast without shrinking with the number of trajectories [1]. At the sequence level the true chain is an ordinary member of the shuffle class: three of the ten fixed shuffles fold more sorted than the primes, and $Z_{\mathrm{seq}} = +1.04$. Inside the sieve model the ordering effect is absent as well (+0.005, $p = 0.68$, run-paired). The exact ideal readout agrees ($z = +0.9$ in the window, Ref. [1]). Where the envelope is flat, the fold does not distinguish the primes from their shuffles.

V. THE POSITIONAL READOUT AND SINGLE CHANNELS

The theorem does not exclude a response to channels that the multiset does not lock, such as the mod-6 doublet of the primes, the strongest Bragg line after parity ($|B_{a/6}|^2 = 0.25$) [10]. We test it with the jitter pair at $N = 4 \times 10^4$ (Table II).

Every jitter contrast is below 0.004, against a shuffle standard deviation of about 0.02, and the kill and keep arms — which differ precisely in whether the channel survives — move the readout by the same small amount. Only the primes separate from their shuffles at all ($z = -2.4$), and that separation is the envelope effect of Refs. [1, 2]. The positional readout does not see single congruence channels.

VI. THE SCATTERING FUNCTION, AND WHY JITTER CANNOT ISOLATE A CHANNEL

The scattering function is the natural candidate for a channel-sensitive readout: a periodic modulation of the stiffness should produce a structure in $S(q)$. Figure 2 shows the exact $S(q)$ of the ideal chain for the primes ($N = 6000$) and the twin primes ($N = 12000$), relative to the shuffle ensemble.

The true family lies inside the shuffle band: its largest deviation is 2.4% for the primes and 1.7% for the twin primes, at most 2.0 standard deviations of the shuffle ensemble. The jitter arms for the primes lie far outside the band, but their displacement does not track the channel. The heavy modulus $q = 6$, whose divisor channels carry Bragg weight $W_3 + W_6 = 1$, moves $S(q)$ no more than the light modulus $q = 7$, with $W_7 = 1/6$ [3]: the mean shift is −3.0% (kill) and −6.1% (keep) at $q = 6$ against −4.1% and −8.0% at $q = 7$. The size of the shift follows the jitter design instead. The variance-matched keep* arm, which perturbs the gap multiset least (Table III), shifts $S(q)$ least (−1.9% and −1.5%), and the three designs rank exactly as their changes of gap variance do. The double difference $[\Delta(6) - \Delta(7)]$, which removes the common mechanical perturbation, is at most 2.2%, 3.1% and 0.8% for the three designs on the primes, with the light modulus contributing the larger shift, and at most 0.6% on the twin primes, whose jitter arms all stay within 2.1% of the shuffle ensemble.

The reason is that the jitter changes the gap multiset, and the polymer is sensitive to composition. Table III audits the three jitter designs on the primes. The keep arm of the original design moves anchors by q or 0, with displacement variance $q^2/4$, three times the kill arm's $(q^2 - 1)/12$, and doubles the change in gap variance. Matching the displacement variance (keep*: $\pm q$ with probability $(q^2-1)/24q^2$ each) does not cure it. The matched arm moves few anchors far while the kill arm moves all anchors a little, and their gap multisets still differ by a factor of four in KS distance. This is not a defect of a particular design. By Theorem 1, any construction that changes a congruence channel must change gap values. A value change that acts only through residues modulo q has its displacements on $q\mathbb{Z}$ in one arm and spread over $\mathbb{Z}/q\mathbb{Z}$ in the other, so the two arms cannot have the same gap statistics.

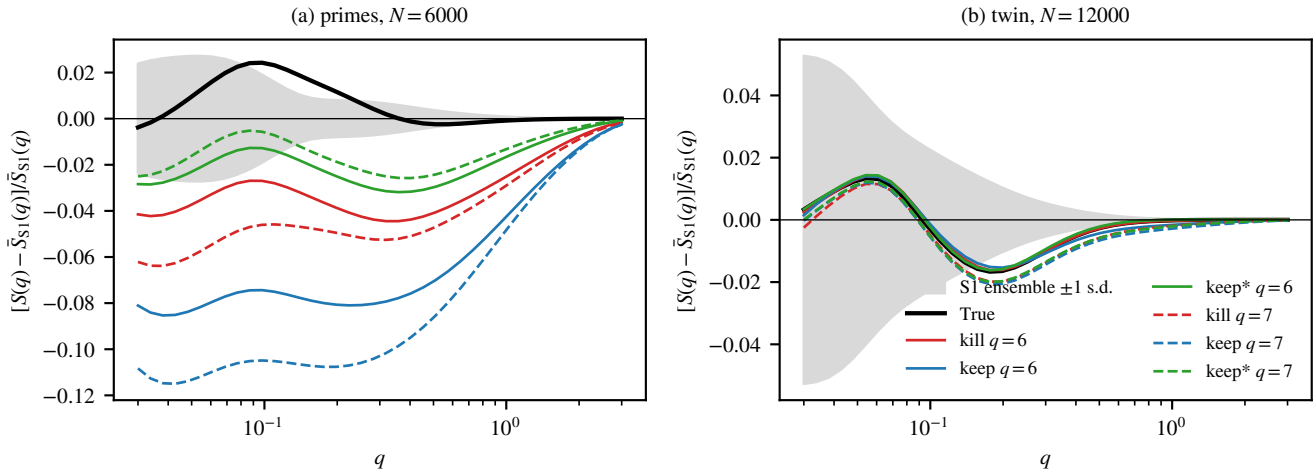


FIG. 2. Exact $S(q)$ of the ideal chain relative to the shuffle ensemble (grey band: ±1 s.d.). Black: true family. Red, blue, green: kill, keep and variance-matched keep (keep*) jitter arms, at the heavy modulus $q = 6$ (solid) and the light modulus $q = 7$ (dashed).

TABLE III. Jitter audit, primes, $N = 4 \times 10^4$, 20 realizations: KS distance of the perturbed gap multiset to the true one, relative change of the gap variance, and Bragg power at a/q (true: 0.249 at $q = 6$, 0.028 at $q = 7$).

q	arm	KS	$\lvert\Delta \mathrm{Var}\, g\rvert/\mathrm{Var}\, g$	Bragg a/q
6	kill	0.083	0.071	2.0×10^{-4}
6	keep	0.071	0.159	0.25
6	keep*	0.021	0.050	0.25
7	kill	0.086	0.094	2.4×10^{-4}
7	keep	0.107	0.155	0.028
7	keep*	0.032	0.048	0.028

VII. COMPOSITION AGAINST SIEVE MODELS

Sections III–VI together show that congruence content cannot be varied in a multiset-preserving design, and cannot be varied without varying composition in a value-perturbing one. A test of congruence structure in the polymer must therefore vary both together, along a family of anchor sets defined arithmetically and otherwise matched. The binned sieve models S_7-G(z) are such a family. Table IV and Fig. 3 compare each family with them at $N = 10^6$.

Composition converges with sieve depth. For every family the composition distance falls monotonically as z grows. The coarse readouts follow it: the mean bond cosine of the primes moves from $z = -63$ against the unsieved model to -8 at $z = 7$. Convergence is not complete at $z = 7$ for the primes, since their sieve continues to $\sqrt{N}$; the polymer registers sieve depth well beyond the first few primes.

The congruence trap is the $z = 3$ layer. For twin and Sophie Germain primes the gaps divisible by 6 rise from 16% in the unsieved model to 100% at $z = 3$, exactly as in the true family. The remaining composition distance then decreases gradually through $z = 5$ and 7. There is no discontinuity at the trap. The single-class confinement is one layer of sieve composition among many, not a separate kind of structure.

The positional readout is blind to all of it. No sieve model is separated from its family by ρ_{id} ($|z| \leq 0.8$), while the S1 shuffle, which keeps composition and destroys the envelope, separates the primes at $z = -7.4$. This is the third independent confirmation that ρ reads the positional envelope [1, 2].

Kernel versus generator. In the Bost–Connes setting, Landau primes are permanently obstructed and twin primes are not [3, Ch. 37, Thm. 2.2]. In the polymer the order is reversed, and the kernel condition plays no role. Twin primes stay distinguishable from their sieve model up to $z = 7$ ($z(\langle\mathcal{L}\rangle) = -6.3$). Landau primes are indistinguishable from every arm on every physical readout ($|z| \leq 2.8$), even the unsieved arm, in which only 46% of the gaps are multiples of 4. The $4\mathbb{Z}$ property of Landau gaps is itself a consequence of parity. Every odd Landau prime has n even, so the gap between any two of them is $(n_2^2+1)-(n_1^2+1) = (n_2-n_1)(n_2+n_1)$, a product of two even numbers, hence in $4\mathbb{Z}$. More to the point, a congruence condition modulo q alters the stiffness composition in proportion to $q/\bar{g}$. It is 0.05 for twin primes at $N = 10^6$ ($q = 6$, $\bar{g} = 122$) and 5×10^{-4} for Landau primes ($q = 4$, $\bar{g} = 8759$). The Landau family has only 112 anchors at this size, so its null result has low power. The ratio $q/\bar{g}$, however, makes the same prediction at any size.

TABLE IV. Prime families against their sieve models at $N = 10^6$, 40 realizations per arm, exact ideal chain. KS: excess Kolmogorov–Smirnov distance between the bead-stiffness distributions of the true family and the arm (the distance between two realizations of the arm subtracted). $z(\cdot)$: true family against the arm, in units of the arm's standard deviation, for the mean bond cosine, the radius of gyration and the positional readout. f_6, f_4: fractions of gaps divisible by 6 and by 4 (true values in the first row of each block).

family (anchors)	arm	KS	$z(\langle\mathcal{L}\rangle)$	$z(R_g^2)$	$z(\rho_{\mathrm{id}})$	f_6	f_4
primes (78 498)	true	—	—	—	—	0.418	0.425
	S1	0.000	−0.9	+6.9	−7.4	0.418	0.425
	$z = 1$	0.093	−63.4	−27.0	−0.8	0.135	0.221
	$z = 3$	0.037	−24.6	−14.2	−0.5	0.434	0.443
	$z = 5$	0.020	−13.9	−13.7	0.0	0.426	0.430
	$z = 7$	0.013	−7.9	−8.2	−0.1	0.422	0.428
twin (8 169)	true	—	—	—	—	1.000	0.473
	$z = 1$	0.053	−11.9	−3.9	−0.3	0.164	0.247
	$z = 3$	0.039	−10.7	−4.3	−0.3	1.000	0.487
	$z = 5$	0.028	−6.6	−3.0	0.0	1.000	0.482
	$z = 7$	0.014	−6.3	−2.8	−0.4	1.000	0.473
Sophie Germain (7 746)	true	—	—	—	—	1.000	0.477
	$z = 1$	0.041	−9.8	−3.6	−0.1	0.164	0.247
	$z = 3$	0.029	−8.9	−2.8	−0.3	1.000	0.486
	$z = 5$	0.017	−6.3	−2.5	+0.1	1.000	0.482
	$z = 7$	0.008	−3.0	−1.6	−0.1	1.000	0.476
Landau (112)	true	—	—	—	—	0.396	0.991
	$z = 1$	< 0	−1.8	+0.8	−0.1	0.228	0.463
	$z = 3$	< 0	−1.3	+1.5	−0.1	0.501	0.996
	$z = 7$	0.004	0.0	+1.8	0.0	0.492	0.995

VIII. DISCUSSION

The five tests leave one picture. A prime-programmed polymer registers arithmetic through two channels only. The first is the positional envelope: how the stiffness composition changes along the chain, which for the primes from the origin is the PNT drift [1, 2]. The second is the composition of the gap multiset, which converges to that of the family as the sieve depth of a model grows. Congruence structure reaches the polymer only as part of the second channel, in proportion to the modulus over the mean gap, and only when the comparison changes composition. No readout we tested responds to the arrangement of residues: not the rank statistic of the collapsed globule, not the positional readout, not the scattering function.

The pattern has a structural parallel in the localized Bost–Connes systems, where the arithmetic of a set of primes lives in the measure class of the equilibrium states and every invariant that tensorizes over the local factors — K-theory, factor type, spectral distance, dynamical entropy — is blind to it [3, Ch. 4]. The polymer readouts share the relevant feature: each is a sum, or a rank statistic of a sum, of local contributions, and the only nonlocal input is a bead's position along the chain. The difference in kind is worth stating. The polymer's equilibrium is a Gibbs measure on conformation space, for one fixed sequence, and its observables are integrals of local functions of the conformation. The KMS measures of the Bost–Connes systems live on a different space, the arithmetic space on which the Hecke semigroup acts, and it is the measure class of those measures that carries the arithmetic. The polymer has no access to that space or to the operator-algebraic structure built on it. Its readouts are expectations of local, additive quantities, the counterpart of the tensorizing invariants rather than of the measure class. We state the parallel as an analogy and not as a theorem. The polymer is not a C*-dynamical system, and the kernel condition that decides uniqueness of the KMS state is a statement about the divergence of character-weighted sums over the family, for which the polymer has no counterpart. The experiment shows that it has none in practice either: the ordering of the families by distance from their sieve models is the opposite of their ordering by obstruction.

The same conclusion bears on the surrogate methodology of the series. A surrogate test of a physical readout can attribute an effect to ordering only through a multiset-preserving null, and Theorem 1 shows that such a null is silent on single-class confinements. A test of congruence structure must change composition, and then a controlled family of composition changes,

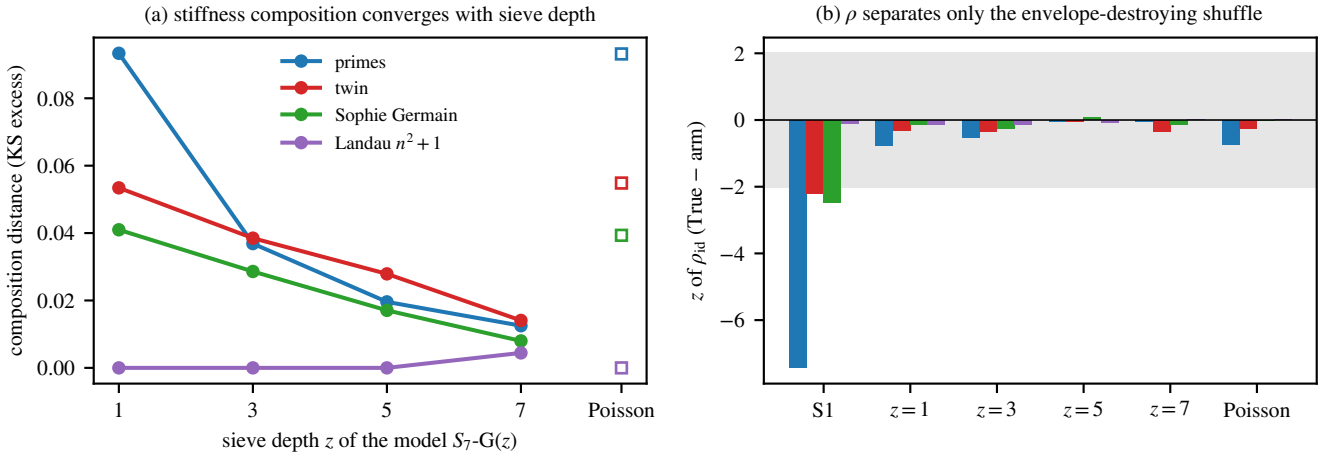


FIG. 3. (a) Composition distance of each family to its sieve model as a function of the sieve depth z (open squares: Poisson; negative values, i.e. closer than two realizations of the arm, are drawn at zero). (b) z-score of the positional readout ρ_{id} of the true family against each arm; grey band $|z| < 2$. Only the S1 shuffle, which destroys the density envelope, separates the primes.

ordered by arithmetic depth, is required. The binned sieve models provide one. We expect the same design to be useful for any physical probe of number-theoretic sequences [11, 12].

Limitations. Most of the results are exact statements about the ideal chain. Excluded volume and collapse enter only through the drift-free window, which uses one size and one protocol. The Landau family is small at accessible sizes. Composition is summarized by a KS distance between stiffness distributions and by a handful of coarse readouts; finer composition statistics might order the families differently. The sieve models are drawn with a fixed bin width of 20 members, so envelope structure on shorter scales is not controlled. Contact-network invariants of the collapsed globule are outside the scope of this paper, but preliminary calculations agree with it. On 400-vertex coarse-grained contact graphs of the production globules, the exact sandpile group (Smith normal form), its 2- and 3-ranks, and the order of the sequence divisor in it do not separate the true chain from shuffled or PNT-only surrogates. The energy of the sequence divisor does separate the true chain from its shuffles, but a PNT-only surrogate matches it, so it reads the low-order positional envelope. After removing Legendre modes up to degree 16, only a marginal difference remains between the true chain and the PNT-only surrogate ($p = 0.043$, uncorrected for multiple testing). These results are the subject of separate work.

IX. CONCLUSION

A polymer whose stiffness follows the gaps of a prime family registers that family's sieve composition and, from the origin, its PNT envelope, and nothing finer. Order contrasts are provably blind to single-class confinement. The positional readout and the scattering function do not respond to single congruence channels. Against sieve models matched in envelope and count, the composition of every family converges monotonically with sieve depth, and the congruence trap is its $z = 3$ layer. The kernel-versus-generator dichotomy that governs the equilibrium states of localized Bost–Connes systems has no polymer counterpart. Congruence conditions act on the fold in proportion to the ratio of modulus to mean gap, which favours dense families such as the twin primes over sparse ones such as the Landau primes.

ACKNOWLEDGMENTS

Code, data, tests and figure scripts are available at https://github.com/Ruqing1963/prime-family-polymer-blindness and archived at doi:10.5281/zenodo.23094095. The exact ideal-chain solver and the Langevin engine are those of the PDTP repository, version 4.0 [1].

[1] R. Chen, Beyond the marginal distribution: prime-gap ordering imprints on a self-avoiding polymer through the small-prime end of the chain (2026), version 4.0, Zenodo,

doi:10.5281/zenodo.22944122.
[2] R. Chen, Collapse, convergence, and the scale of the prime-gap imprint on polymer dynamics (2026), Zenodo, doi:10.5281/zenodo.23076433.
[3] R. Chen, Localized Bost–Connes systems (2026), Zenodo, doi:10.5281/zenodo.22883179.
[4] J.-B. Bost and A. Connes, Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory, Selecta Math. (N.S.) **1**, 411 (1995).
[5] J. D. Weeks, D. Chandler, and H. C. Andersen, Role of repulsive forces in determining the equilibrium structure of simple liquids, J. Chem. Phys. **54**, 5237 (1971).
[6] D. L. Ermak and J. A. McCammon, Brownian dynamics with hydrodynamic interactions, J. Chem. Phys. **69**, 1352 (1978).
[7] H. Cramér, On the order of magnitude of the difference between consecutive prime numbers, Acta Arith. **2**, 23 (1936).
[8] A. Granville, Harald Cramér and the distribution of prime numbers, Scand. Actuar. J. **1995**, 12 (1995).
[9] T. Schreiber and A. Schmitz, Improved surrogate data for nonlinearity tests, Phys. Rev. Lett. **77**, 635 (1996).
[10] S. Torquato, G. Zhang, and M. de Courcy-Ireland, Uncovering multiscale order in the prime numbers via scattering, J. Stat. Mech. , 093401 (2018).
[11] T. Odijk, (2019), arXiv:1904.09693.
[12] A. R. Khokhlov and P. G. Khalatur, Conformation-dependent sequence design (engineering) of AB copolymers, Phys. Rev. Lett. **82**, 3456 (1999).